DESKTOP PC ASSEMBLY AND DISASSEMBLY – PRACTICAL ASSIGNMENT

Practical 1: Identify the Main Components Inside a Desktop PC

Aim

To open a desktop computer cabinet and identify the main internal hardware components.

Main Components

No.	Component	Short Description
1	**Motherboard**	The main circuit board of the computer. It connects the CPU, RAM, storage, expansion cards and other components.
2	**CPU / Processor**	The main processing unit of the computer. It executes instructions and performs calculations required by software.
3	**RAM**	Temporary memory used by the CPU to store data and programs that are currently being used.
4	**GPU / Graphics Card**	Processes graphics and sends visual output to the monitor. It may be integrated or a separate dedicated card.
5	**HDD**	A magnetic storage device used to permanently store the operating system, applications and files.
6	**SSD**	A flash-based storage device that provides faster data access than a traditional HDD.
7	**Power Supply Unit (PSU)**	Converts AC electricity from the wall outlet into the DC power required by computer components.
8	**CPU Cooler**	Removes heat from the processor using a heatsink and fan or another cooling system.
9	**Case Fans**	Move air through the cabinet to remove hot air and bring in cooler air.
10	**CMOS Battery**	Small battery on the motherboard that helps maintain BIOS/UEFI settings and system time when the PC is powered off.
11	**SATA Cables**	Used to connect SATA storage devices such as HDDs and SATA SSDs to the motherboard.
12	**M.2 SSD**	A compact storage device installed directly into an M.2 slot on compatible motherboards.
13	**Expansion Slots**	Slots such as PCIe that allow additional cards, including graphics, sound or network cards, to be installed.

No.	Component	Short Description
14	**Optical Drive**	A drive used to read or write CDs/DVDs. It is optional and is not present in many modern PCs.
15	**Front-Panel Connectors**	Small connectors that connect the power button, reset button, LEDs and front-panel ports to the motherboard.

Observation

After opening the cabinet, different hardware components were visible. Each component had a specific role in processing, storing, powering or cooling the computer.

Result

The major internal components of a desktop PC were successfully identified and their functions were understood.

Practical 2: Disassemble a Desktop PC and Document Each Step

Aim

To safely disassemble a desktop PC and document the procedure using photographs or labeled diagrams.

Requirements

- Desktop PC
- Screwdriver set
- Clean working table
- Small container for screws
- Anti-static wrist strap, if available
- Camera/mobile phone for photographs

Safety Precautions

Before starting, shut down the computer completely and disconnect it from electrical power.

Step 1: Shut Down the PC

Save all work and shut down the operating system normally.

Caption:
The computer was completely shut down before starting the disassembly process.

Photo/Diagram:

[Insert Photo 1 here]

Step 2: Disconnect the Power Cable

Remove the power cable from the PSU and disconnect other external cables such as the monitor, keyboard and mouse.

Caption:
All external cables were disconnected to isolate the computer from external power and peripherals.

Photo/Diagram:

[Insert Photo 2 here]

Step 3: Place the PC on a Suitable Surface

Place the cabinet on a clean, stable and non-conductive working surface.

Caption:
The cabinet was placed on a clean working surface to provide enough space for safe disassembly.

Photo/Diagram:

[Insert Photo 3 here]

Step 4: Remove the Side Panel

Remove the screws holding the side panel and carefully slide or remove the panel according to the cabinet design.

Caption:
The side panel was removed to access the internal components.

Photo/Diagram:

[Insert Photo 4 here]

Step 5: Take an Initial Internal Photo

Before disconnecting anything, take a photograph of the complete internal arrangement.

This is useful as a reference during reassembly.

Caption:
An initial photograph was taken to remember the location and routing of internal cables.

Photo:

[Insert Photo 5 here]

Step 6: Disconnect Power Cables

Disconnect the required power cables from components such as:

- Motherboard

- CPU
- GPU
- Storage drives

Caption:
Internal power cables were disconnected carefully by holding the connectors rather than pulling the wires.

Photo/Diagram:

[Insert Photo 6 here]

Step 7: Remove RAM

Release the clips on the RAM slot and carefully remove the RAM module by holding its edges.

Caption:
The RAM module was removed after releasing the motherboard slot clips.

Photo/Diagram:

[Insert Photo 7 here]

Step 8: Remove Storage Drive

Disconnect the required SATA/power cables and remove the HDD or SSD from its mounting position.

For an M.2 SSD, remove the retaining screw and carefully remove the module.

Caption:
The storage drive was disconnected and removed from its mounting position.

Photo/Diagram:

[Insert Photo 8 here]

Step 9: Remove Graphics Card

If a dedicated GPU is installed, disconnect its power cable if present.

Remove the screw securing the GPU and release the PCIe slot latch before carefully removing the card.

Caption:
The dedicated graphics card was removed from the PCIe expansion slot.

Photo/Diagram:

[Insert Photo 9 here]

Step 10: Disconnect Front-Panel and Other Cables

Disconnect connectors for:

- Power button
- Reset button
- Front USB
- Front audio
- Case fans

Caption:
Front-panel and other motherboard connectors were disconnected carefully.

Photo/Diagram:

[Insert Photo 10 here]

Step 11: Remove the Motherboard

After all required connections and expansion cards have been removed, remove the screws securing the motherboard to the cabinet.

Carefully lift the motherboard out of the case.

Caption:
The motherboard was carefully removed after all connected cables and mounting screws were disconnected.

Photo/Diagram:

[Insert Photo 11 here]

Disassembly Observation

The desktop PC was disassembled in a controlled sequence. Power was disconnected first, followed by removal of the cabinet panel, cables, RAM, storage devices, expansion cards and motherboard.

Result

The main components of the desktop PC were successfully identified and the disassembly process was documented using photographs/diagrams.

Practical 3: Reassemble the Desktop PC

Aim

To reinstall the components of a desktop PC in the correct order and verify that the computer starts successfully.

Reassembly Procedure

Step 1: Install the Motherboard

Place the motherboard correctly inside the cabinet and align it with the case mounting points and rear I/O opening.

Secure it using the appropriate screws.

Caption:
The motherboard was installed first so that other components could be connected to their correct motherboard slots and headers.

Photo:

[Insert Photo 12 here]

Step 2: Connect the Power Supply Cables

Connect the main motherboard power connector and CPU power connector.

If required, connect GPU power cables and other component power cables.

Caption:
The required PSU cables were connected to provide power to the motherboard and components.

Photo:

[Insert Photo 13 here]

Step 3: Install RAM

Align the RAM notch with the slot key and press the module down until the retaining clips lock into place.

Caption:
The RAM module was installed in the correct memory slot and secured using the slot clips.

Photo:

[Insert Photo 14 here]

Step 4: Install the Storage Drive

Install the HDD/SSD in its mounting location.

Connect the required SATA data and power cables, or install an M.2 SSD into its M.2 slot.

Caption:
The storage drive was installed and connected so that the operating system and user data could be accessed.

Photo:

[Insert Photo 15 here]

Step 5: Install the Graphics Card

Insert the dedicated GPU into the appropriate PCIe slot.

Secure it to the case and connect its power cable if required.

Caption:
The graphics card was installed into the PCIe slot and secured to the cabinet.

Photo:

[Insert Photo 16 here]

Step 6: Connect Front-Panel Cables

Reconnect the power switch, reset switch, power LED, USB and front-audio connectors according to the motherboard manual.

Caption:
Front-panel connectors were reconnected so that the cabinet buttons and front ports would function correctly.

Photo:

[Insert Photo 17 here]

Step 7: Connect Case Fans

Reconnect the case fans to the appropriate motherboard fan headers or fan controller.

Caption:
The case fans were connected to maintain proper airflow and cooling.

Photo:

[Insert Photo 18 here]

Step 8: Check All Connections

Before closing the cabinet, check:

- RAM is locked correctly.
- GPU is seated properly.
- Storage cables are connected.
- Motherboard power cable is connected.
- CPU power cable is connected.

- Front-panel connectors are connected.
- No loose screws or cables are present.
- Fans can rotate freely.

Caption:
A final inspection was performed before powering on the computer.

Step 9: Close the Cabinet

Replace and secure the side panel.

Photo:

[Insert Photo 19 here]

Step 10: Reconnect External Cables

Reconnect:

- Power cable
- Monitor cable
- Keyboard
- Mouse
- Network cable, if required

Step 11: Power On and Test

Turn on the computer.

Check whether:

- The fans start spinning.
- The display appears.
- The system enters BIOS/UEFI or Windows.
- RAM is detected.
- Storage is detected.

Caption:
The PC was powered on after reassembly and checked for successful operation.

Photo:

[Insert Photo 20 here]

Why Does the Order Matter?

The order matters because some components depend on others for proper installation and access.

For example, the motherboard should be securely mounted before connecting many internal components. RAM and expansion cards must be inserted into their correct slots. Power and data cables must be connected correctly before testing the system.

Following a proper sequence also reduces the chance of damaging components or forgetting an important connection.

Result

The desktop PC was successfully reassembled and tested. The system booted successfully after the components and cables were correctly installed.

Practical 4: Safety Precautions and Tools Checklist

A. Safety Checklist

✓ Safety Precaution	Why It Is Important
☐ Shut down the PC	Prevents electrical and hardware damage.
☐ Disconnect power cable	Prevents electric shock and accidental powering.
☐ Disconnect peripherals	Provides a safer and cleaner working area.
☐ Work on a clean table	Prevents dust and small objects from entering components.
☐ Use anti-static protection	Reduces the risk of electrostatic discharge damaging components.
☐ Hold components by the edges	Prevents contact with sensitive electrical contacts.
☐ Do not force connectors	Prevents damage to ports and connectors.
☐ Keep screws organized	Prevents lost screws and incorrect screw placement.
☐ Take photos before disconnecting	Provides a reference during reassembly.
☐ Follow the motherboard manual	Helps identify connectors and installation positions.
☐ Keep liquids away	Prevents short circuits and component damage.
☐ Check connections before power-on	Reduces the chance of incorrect connections.

B. Tools Required

Tool	Purpose
Phillips Screwdriver	Used to remove and install common PC screws.
Flat-head Screwdriver	Useful for certain screws or clips when appropriate.
Anti-static Wrist Strap	Helps protect components from electrostatic discharge.
Small Screw Container	Keeps screws organized during disassembly.
Flashlight	Helps see connectors and components inside the cabinet.
Cable Ties	Used to organize cables after reassembly.
Mobile Phone/Camera	Used to photograph each disassembly and reassembly step.
Clean Working Surface	Provides a safe area for working with components.

Final Checklist Before Switching On

☐ Motherboard securely installed
☐ RAM properly locked
☐ CPU cooler connected
☐ GPU properly seated
☐ GPU power cable connected if required
☐ Storage drive installed
☐ SATA/M.2 connections checked
☐ 24-pin motherboard power connected
☐ CPU power connector connected
☐ Front-panel connectors connected
☐ Case fans connected
☐ No loose screws inside cabinet
☐ No cables touching fan blades
☐ Side panel installed
☐ Monitor and power cable connected

Overall Conclusion

This practical provided hands-on knowledge of desktop PC hardware. The major internal components were identified and their functions were studied. The PC was then disassembled carefully by following proper safety procedures and documenting each step.

After disassembly, the components were reinstalled in the correct sequence. The final connections were checked and the system was powered on to verify successful reassembly.

Learning Outcomes

After completing this practical, the student can:

1. Identify major components inside a desktop PC.
2. Explain the basic function of each component.
3. Safely open and disassemble a desktop cabinet.
4. Identify power, data and front-panel connections.
5. Remove and reinstall RAM and storage devices.
6. Understand the correct sequence for PC assembly.
7. Use basic tools required for PC maintenance.
8. Apply anti-static and electrical safety precautions.
9. Troubleshoot basic connection problems after reassembly.
10. Verify whether a reassembled PC boots successfully.
